

# Numerical Experiments for Lévy Score-Corrected Compound-Poisson Equilibrium Sampling

## Abstract

We evaluate Lévy score-corrected compound-Poisson (LSC-CP) sampling on four multimodal Boltzmann model systems: a one-dimensional double well, a planar forty-mode Gaussian mixture, a depth-balanced three-well Müller–Brown landscape embedded in ten dimensions, and a coupled  $\phi^4$  chain in twenty-four dimensions. Across the reported horizons, LSC-CP markedly reduces terminal basin-population error relative to local samplers and raw compound-Poisson dynamics, and it yields nonzero post-settling effective sample sizes for every basin indicator. The MoG40 global  $W_2$  and MMD discrepancies remain above their finite-sample reference bands, so the results do not imply that every distributional metric reaches its sampling floor. The implementation uses finite chord quadrature, tamed Euler updates, an exponent cap, and projection onto a finite computational box. These practical stabilizations control overflow and large updates and improve numerical robustness in the stiff score regime, but they define an approximate finite-step kernel. Statistically resolved potential-energy biases are therefore reported explicitly. The evidence supports accelerated interbasin exploration and accurate population recovery with observable-dependent numerical bias, rather than exact finite-step Boltzmann sampling. No physical kinetics is inferred.

## 1 Overview and compared methods

All experiments target a Boltzmann law

$$\pi_\beta(x) = Z_\beta^{-1} e^{-\beta V(x)}, \quad x \in \mathbb{R}^d. \quad (1)$$

The sampler of interest augments overdamped Langevin dynamics with a finite compound-Poisson jump process of measure  $\nu(dr) = \lambda\rho(dr)$  and adds the stationary Lévy-score drift  $S_{\nu,\beta}$  so that the combined generator

$$A_{\nu,\beta}f = [-\nabla V + S_{\nu,\beta}] \cdot \nabla f + \beta^{-1} \Delta f + J_\nu f \quad (2)$$

is weakly stationary for  $\pi_\beta$  (Appendix A). The practical estimators evaluate the score integral by deterministic chord quadrature where affordable and by *paired* random-bank estimators in higher dimension: the realized jump displacements and the Lévy-score drift of one step are generated from the same random finite measure. This pairing aligns the drift and jump measures before the remaining finite chord quadrature is applied (Appendix A.1). Figures label the implemented estimator simply LSC-CP. The reported variants are: deterministic shell quadrature (double well), a semi-analytic score with angular quadrature (MoG40), the paired single-atom estimator LSC-CP-RA (double well and MoG40, as an estimator-agreement check), and the paired multi-atom estimator LSC-CP-MA (Müller–Brown and  $\phi^4$ ).

Baselines are the unadjusted Langevin algorithm (ULA), Metropolis adjusted Langevin (MALA), the BAOAB underdamped integrator (BAOAB), an  $\alpha$ -stable Lévy-flight analogue (FLA), and parallel tempering (PT) with an automatically tuned ladder and fully counted replica cost. Raw compound

Poisson (CP), i.e. the same jump bank *without* the score correction, is included as a stationarity-defect ablation: its invariant law is not  $\pi_\beta$ , so it is a control for the value of the correction, not a fair efficiency baseline. All methods share per-seed initial ensembles. In the double-well and MoG40 full-law experiments, CP and deterministic/closed-form LSC-CP additionally share the jump random stream pathwise. In the Müller–Brown and  $\phi^4$  experiments, raw full-law CP and LSC-CP-MA match the marginal jump geometry, weights, and intensity, but do not use the same realized random banks and hence are not pathwise paired.

The implemented scheme uses finite chord quadrature, log-domain accumulation with an exponent cap, tamed Euler drift updates, and projection onto a finite computational domain to stabilize evaluation in the stiff-score regime. These operations define the finite-step algorithm evaluated below. The stationarity identity applies to the ideal continuous-time generator with the exact, uncapped Lévy-score field on an unbounded state space. We assess the equilibrium fidelity of the discretization through timestep and quadrature sensitivity checks, observable biases, and cap/projection activation rates. Passage statistics are interpreted solely as algorithmic mixing diagnostics; no physical reaction kinetics is inferred.

## 2 Benchmark systems

|                        | Double well             | MoG40                              | Depth-balanced MB      | Coupled $\phi^4$        |
|------------------------|-------------------------|------------------------------------|------------------------|-------------------------|
| dimension $d$          | 1                       | 2                                  | 10                     | 24                      |
| $\beta$                | 8                       | 8                                  | 24                     | 8                       |
| particles $N$          | 16,000                  | 2,500                              | 2,000                  | 1,000                   |
| horizon $T$            | 100                     | 100                                | 200                    | 100                     |
| $\Delta t$ (used)      | 0.005                   | 0.010                              | 0.005                  | 0.002                   |
| steps                  | 20,000                  | 10,000                             | 40,000                 | 50,000                  |
| seeds                  | 24                      | 24                                 | 16                     | 16                      |
| jump rate $\lambda$    | 1                       | 1                                  | 1                      | 1                       |
| score quadrature       | $q_\theta=16, q_\rho=4$ | analytic $\theta, \rho; m_\phi=16$ | $q_\theta=8, q_\rho=4$ | $q_\theta=32, q_\rho=4$ |
| score estimator        | deterministic + RA      | semi-analytic + RA                 | MA                     | MA                      |
| weak-identity residual | $1.5 \times 10^{-7}$    | $3.0 \times 10^{-8}$               | $5.0 \times 10^{-7}$   | $2.0 \times 10^{-7}$    |

Table 1: Reported numerical configurations and uncapped weak drift–jump identity residuals (diagnostic tolerance  $10^{-6}$ ; Appendix E). The residual measures generator-level drift–jump cancellation for a finite family of test functions. Full model, bank, and ladder specifications are in Appendix B.

**Double well** ( $d = 1$ ).  $V(x) = (x^2 - 1)^2$  at  $\beta = 8$ : two equal wells at  $\pm 1$  separated by a dimensionless barrier  $\beta\Delta V = 8$ . The jump bank is a symmetric shell at  $\pm 2$  with radial half-width 0.2. This is the transparent two-state test of target recovery and interwell communication.

**MoG40** ( $d = 2$ ).  $\pi_\beta$  is an equal-weight mixture of forty unit-covariance Gaussians;  $V(x) = -\beta^{-1} \log \sum_k \exp[-\|x - \mu_k\|^2/2]$ . The jump law is deliberately generic: radius uniform on  $[4, 15]$  (set from the nearest-neighbour distance histogram alone), direction uniform on the circle. Neither PT nor LSC-CP receives the mode locations. This is the many-basin stress test.

**Depth-balanced Müller–Brown** ( $d = 10$ ). A Müller–Brown-type three-well landscape whose well depths are retuned to be equal, evaluated in a two-dimensional latent plane and embedded in

ten dimensions by a fixed linear map with Gaussian auxiliary coordinates ( $\sigma_{\text{aux}} = 0.4$ ), at  $\beta = 24$ . The equal depths separate multimodality from metastability: the channels retain dimensionless barriers  $\beta b_{AB} \simeq 11.1$  and  $\beta b_{BC} \simeq 4.0$  while the equilibrium masses stay comparable. The bank has four directed relay atoms  $A \leftrightarrow B \leftrightarrow C$  (no direct  $A$ – $C$  atom), so transport must use the relay through  $B$ . This tests curved, nonuniform basins and a designed jump graph.

**Coupled  $\phi^4$  chain ( $d = 24$ ).** Twelve periodic two-component sites,  $V(q) = \frac{\kappa}{2\delta} \sum_i \|q_{i+1} - q_i\|^2 + \delta \sum_i W(q_i)$  with  $\kappa = 2.5$ ,  $\delta = 1/12$ ,  $\beta = 8$ , and a tilted double-double-well site potential  $W$  (Appendix B). The target has four coherent phases; the bank has eight directed edge atoms along the phase square (diagonals excluded because their coherent chords cross the field-zero hilltop). The primary collective variable is the spatial mean  $\bar{q} = \frac{1}{12} \sum_i q_i \in \mathbb{R}^2$ . This is the many-body example: a collective transition in a coupled field.

The four systems form a hierarchy of equilibrium questions—two-state balance, many-mode occupancy, curved-basin relay transport, and a collective field transition—rather than four independent leaderboard tasks.

### 3 Protocol, references, and numerical parameter selection

**Run protocol.** For each system all seeds are batched into one ensemble per method with per-seed initial blocks; every method starts from the identical localized per-seed ensemble. We used dyadic timestep sensitivity checks based on terminal  $W_2$ , basin TV, MMD, and energy-moment-consistency diagnostics for the  $\pi_\beta$ -targeting methods. The non-targeting FLA and raw CP chains were recorded but did not determine the timestep. Score quadrature was selected analogously, with the additional uncapped weak-identity residual. These checks guided numerical parameter selection, but they did not include post-settling mean energy or collective-variable bias and therefore do not establish convergence of the invariant measure of the stabilized chain.

**References and floors.** The double well uses a dense-grid inverse-CDF reference; MoG40 an exact mixture draw; Müller–Brown a fine two-dimensional latent grid combined with exact Gaussian auxiliaries;  $\phi^4$  a Laplace-mixture proposal corrected by self-normalized importance weights (direct SNIS for basin masses, energy moments, CV means, and the free-energy surface; finite sampling-importance-resampling only where unweighted clouds are required). The  $\phi^4$  weight diagnostics for the reference are:  $2 \times 10^5$  proposals, weight ESS 41,182 (fraction 0.206), maximum normalized weight  $3.6 \times 10^{-3}$  (Appendix D). A separate  $t = 300$  parallel-tempering check gave phase masses (0.365, 0.204, 0.234, 0.197), whose TV distance from the SNIS masses is 0.042. Consequently, differences below this scale in the  $\phi^4$  basin metrics are reference sensitive. For every metric, a finite-sample *floor* is estimated by comparing two independent reference draws of the same size; “at the floor” below means within the floor band mean + 3 sd.

**Stability diagnostics.** We monitored nonfinite states, score-cap and state-box projection frequencies, jump-boundary contact per applied jump, and basin-map-domain escape. The uncapped weak-identity residual was evaluated on a domain extending at least one maximum jump beyond the high-probability region. The measured values and domain-coverage rationale appear in Appendix E. These quantities diagnose numerical robustness and exposure to the stabilizations; small activation frequencies do not prove that the finite-step kernel has invariant law  $\pi_\beta$ .

## 4 Equilibrium recovery

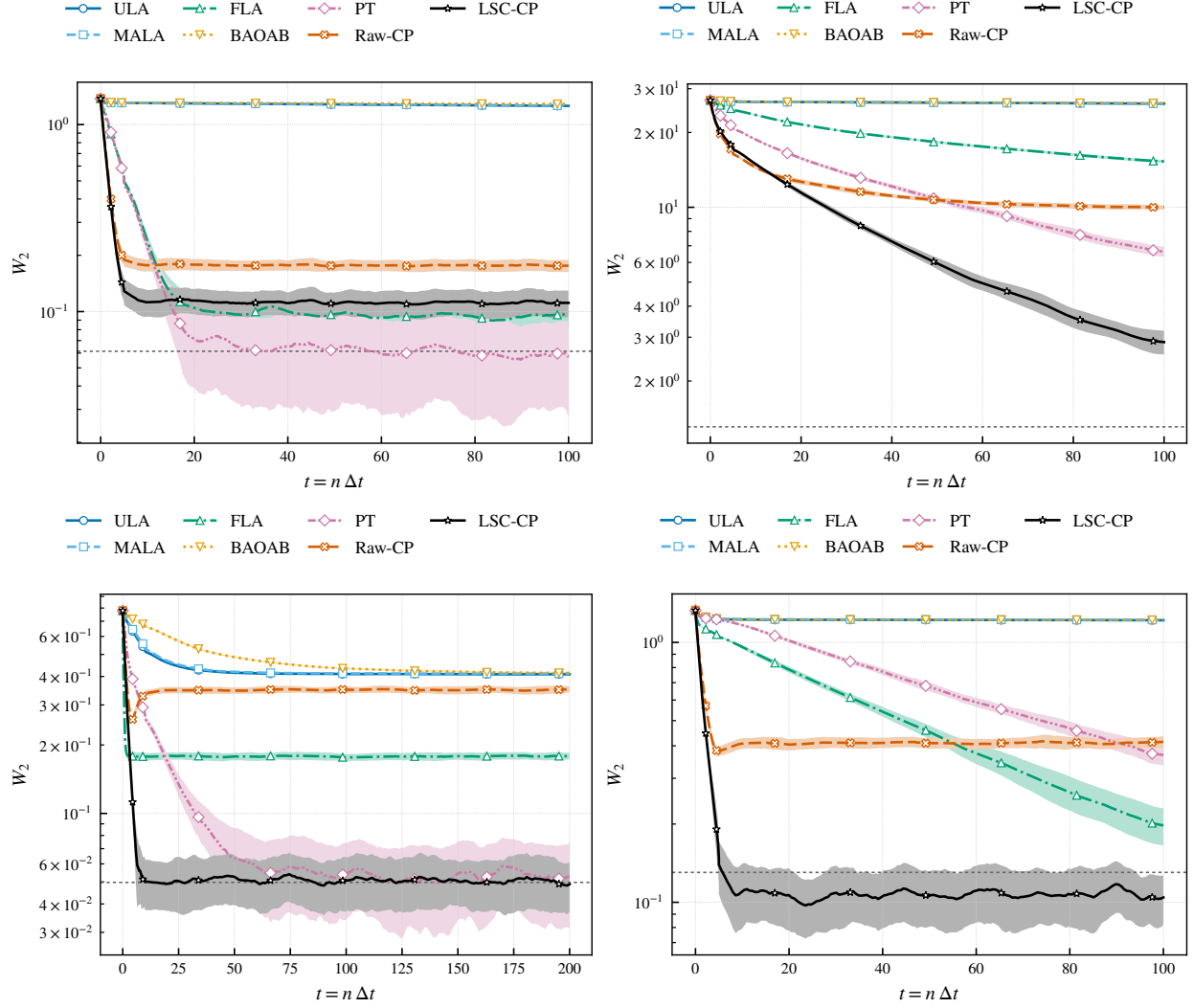


Figure 1: Ensemble relaxation:  $W_2$  to the reference versus simulation time  $t$  from the shared localized initial ensembles (seed-band envelopes). Top: double well, MoG40; bottom: Müller–Brown 10D, coupled  $\phi^4$ .

Four observations organize Table 2 and Figure 1.

(i) *Local dynamics remain strongly metastable.* On every system, the terminal ULA, MALA, and BAOAB ensembles remain strongly localized relative to the target (basin TV 0.31–0.93, basin KL 1.1–4.2). Occasional label changes occur in some of the post-settling MoG40 and Müller–Brown traces, but they are too sparse to recover all basin populations. This quantifies the metastability that the nonlocal methods are intended to address.

(ii) *LSC-CP improves basin recovery, with metric-dependent residual error.* The reported LSC-CP estimator attains basin TV and KL at or within the finite-sample floor band on the double well, Müller–Brown, and  $\phi^4$ ; the last comparison is subject to the independent-reference uncertainty noted in Section 3. Its MoG40 basin TV, 0.064, is also within the upper reference band,  $0.051 + 3(0.006) = 0.069$ . The corresponding global  $W_2 = 2.83$  exceeds its upper reference band of

| Method                                                                                            | $W_2$         | $TV_{\text{basin}}$ | $RMSE_F/k_B T$ | $D_{\text{KL}}(p^* \ \tilde{p})$ | FES outside |
|---------------------------------------------------------------------------------------------------|---------------|---------------------|----------------|----------------------------------|-------------|
| <i>Double well</i> (floors: $W_2$ $0.061 \pm 0.033$ , TV $0.0032 \pm 0.0017$ )                    |               |                     |                |                                  |             |
| ULA                                                                                               | 1.26 (0.003)  | 0.471 (0.001)       | 1.77 (0.02)    | 1.09 (0.02)                      | 0           |
| MALA                                                                                              | 1.26 (0.003)  | 0.473 (0.001)       | 1.79 (0.02)    | 1.12 (0.02)                      | 0           |
| BAOAB                                                                                             | 1.29 (0.002)  | 0.488 (0.001)       | 2.23 (0.04)    | 1.54 (0.04)                      | 0           |
| FLA                                                                                               | 0.096 (0.018) | 0.0033 (0.003)      | 0.302 (0.008)  | $3.4 \times 10^{-5}$             | 0.003       |
| PT                                                                                                | 0.064 (0.032) | 0.0033 (0.002)      | 0.050 (0.009)  | $3.2 \times 10^{-5}$             | 0           |
| CP                                                                                                | 0.176 (0.013) | 0.0036 (0.003)      | 0.157 (0.006)  | $4.2 \times 10^{-5}$             | 0.029       |
| LSC-CP                                                                                            | 0.111 (0.017) | 0.0036 (0.003)      | 0.168 (0.009)  | $4.4 \times 10^{-5}$             | 0.005       |
| LSC-CP-RA                                                                                         | 0.131 (0.017) | 0.0031 (0.003)      | 0.180 (0.010)  | $3.2 \times 10^{-5}$             | 0.008       |
| <i>MoG40</i> (floors: $W_2$ $1.31 \pm 0.28$ , TV $0.051 \pm 0.006$ )                              |               |                     |                |                                  |             |
| ULA                                                                                               | 26.1 (0.02)   | 0.916 (0.003)       | 1.81 (0.01)    | 4.20 (0.02)                      | 0           |
| MALA                                                                                              | 26.1 (0.03)   | 0.917 (0.003)       | 1.81 (0.01)    | 4.21 (0.01)                      | 0           |
| BAOAB                                                                                             | 26.2 (0.03)   | 0.930 (0.005)       | 1.77 (0.01)    | 4.24 (0.02)                      | 0           |
| FLA                                                                                               | 15.2 (0.21)   | 0.436 (0.007)       | 1.37 (0.06)    | 0.743 (0.05)                     | 0.001       |
| PT                                                                                                | 6.54 (0.34)   | 0.141 (0.007)       | 0.350 (0.02)   | 0.055 (0.005)                    | 0           |
| CP                                                                                                | 9.98 (0.19)   | 0.286 (0.009)       | 0.680 (0.04)   | 0.267 (0.013)                    | 0.254       |
| LSC-CP                                                                                            | 2.83 (0.34)   | 0.064 (0.008)       | 0.190 (0.02)   | 0.012 (0.003)                    | 0.001       |
| LSC-CP-RA                                                                                         | 2.65 (0.26)   | 0.103 (0.009)       | 0.280 (0.02)   | 0.036 (0.006)                    | 0.005       |
| <i>Depth-balanced MB, 10D</i> (floors: $W_2$ $0.050 \pm 0.020$ , TV $0.011 \pm 0.007$ )           |               |                     |                |                                  |             |
| ULA                                                                                               | 0.409 (0.003) | 0.307 (0.002)       | 2.27 (0.23)    | 1.17 (0.14)                      | 0           |
| MALA                                                                                              | 0.411 (0.003) | 0.308 (0.001)       | 2.32 (0.18)    | 1.22 (0.16)                      | 0           |
| BAOAB                                                                                             | 0.415 (0.003) | 0.309 (0.001)       | 2.50 (0.17)    | 1.52 (0.21)                      | 0           |
| FLA                                                                                               | 0.179 (0.006) | 0.119 (0.008)       | 0.941 (0.04)   | 0.048 (0.007)                    | 0.039       |
| PT                                                                                                | 0.056 (0.021) | 0.013 (0.008)       | 0.132 (0.02)   | $5.7 \times 10^{-4}$             | 0           |
| CP                                                                                                | 0.353 (0.008) | 0.135 (0.006)       | 0.267 (0.02)   | 0.047 (0.005)                    | 0.172       |
| LSC-CP-MA                                                                                         | 0.047 (0.009) | 0.014 (0.006)       | 0.136 (0.010)  | $5.8 \times 10^{-4}$             | 0.006       |
| <i>Coupled <math>\phi^4</math>, 24D</i> (floors: $W_2$ $0.130 \pm 0.035$ , TV $0.024 \pm 0.010$ ) |               |                     |                |                                  |             |
| ULA                                                                                               | 1.22 (0.004)  | 0.671 (0.003)       | 2.59 (0.09)    | 3.07 (0.31)                      | 0           |
| MALA                                                                                              | 1.22 (0.004)  | 0.673 (0.002)       | 2.63 (0.05)    | 3.13 (0.17)                      | 0           |
| BAOAB                                                                                             | 1.22 (0.006)  | 0.673 (0.002)       | 2.64 (0.05)    | 3.14 (0.22)                      | 0           |
| FLA                                                                                               | 0.194 (0.032) | 0.060 (0.013)       | 0.220 (0.03)   | 0.011 (0.004)                    | $10^{-4}$   |
| PT                                                                                                | 0.364 (0.028) | 0.119 (0.015)       | 0.315 (0.04)   | 0.039 (0.008)                    | 0           |
| CP                                                                                                | 0.410 (0.021) | 0.077 (0.016)       | 0.482 (0.05)   | 0.017 (0.006)                    | 0.209       |
| LSC-CP-MA                                                                                         | 0.100 (0.028) | 0.022 (0.011)       | 0.151 (0.03)   | 0.0015 (0.001)                   | 0.003       |

Table 2: Terminal metrics at  $t = T$ , mean (std) over seeds.  $W_2$ : sliced (exact in 1D) Wasserstein-2 against the reference in metric space;  $TV_{\text{basin}}$ : total variation of basin occupancy;  $RMSE_F$ : additive-aligned free-energy-surface RMSE in  $k_B T$  on the two-dimensional collective variables (one-dimensional for the double well);  $D_{\text{KL}}$ : target-to-empirical basin KL with Jeffreys pseudocount; FES outside: empirical mass outside the frozen reference FES grid. Metric definitions in Appendix C.

approximately 2.15, however, and  $\text{MMD} = 0.0482$  exceeds its upper band of approximately 0.0317. On MoG40, the raw-jump and PT basin-TV errors are respectively 4.5 and 2.2 times the LSC-CP value, while LSC-CP-RA is 1.6 times larger. Parallel tempering matches LSC-CP on the double well and Müller–Brown but has larger basin KL on MoG40 (0.055 versus 0.012) and  $\phi^4$  (0.039 versus 0.0015) at full replica cost. The latter distinction is reference-sensitive below the 0.042 scale. The  $\alpha$ -stable flight is competitive on the double well but shows much larger discrepancies elsewhere (e.g., MB free-energy RMSE  $0.94 k_B T$ ).

(iii) *The score correction substantially reduces the raw-jump distortion.* In the double-well and MoG40 full-law runs, raw CP and LSC-CP share jump streams pathwise. The MB and  $\phi^4$  comparisons instead match the marginal bank geometry, weights, and intensity but are not pathwise paired. Raw CP retains basin TV 0.286/0.135/0.077 on MoG40/MB/ $\phi^4$ , and 17–25% of its terminal mass lies outside the reference free-energy grid on the three multivariate systems, versus  $\leq 0.6\%$  for LSC-CP. Raw CP also contacts the finite box on 2.5%, 6.7%, and 0.4% of applied jumps on these systems (Appendix E); this is a numerical-box confound, especially for MoG40 and MB, rather than an intrinsic component of the stationary defect. In one  $\phi^4$  sensitivity comparison, widening the box from  $[-2, 2]^{24}$  to  $[-5, 5]^{24}$  changes raw-CP  $W_2$  from 0.28 to 0.41, whereas LSC-CP-MA changes from 0.099 to 0.100. This single check supports robustness to that particular box change but does not establish general box-size independence.

(iv) *Random-bank and deterministic estimators are qualitatively consistent.* Where both are run (double well and MoG40), the paired single-atom estimator preserves the qualitative method ordering and gives comparable terminal errors on several metrics. It is not numerically equivalent on every metric: for example, the MoG40 basin KL is 0.036 versus 0.012 (a factor of about three), and the outside-grid mass is about four times larger. The comparison therefore provides an estimator-sensitivity check, not a proof of equivalence or a validation of the multi-atom approximation.

## 5 Post-settling time-series diagnostics

Terminal ensemble metrics quantify the law at time  $T$ ; they do not measure the precision of time averages. For that purpose, small independent chains (4 seeds  $\times$  8 chains, 1000 uniformly spaced draws over one additional horizon) are initialized from the system reference, evolved for one full settling horizon, and then analyzed for basin indicators, energy, and collective variables. None is claimed to be an exact stationary draw of the discrete kernel, so the values below are post-settling diagnostics of the numerical chains (Appendix C). A constant basin indicator contributes zero ESS by construction. Raw CP and FLA are excluded because their invariant laws are not  $\pi_\beta$ .

The LSC energy MCSEs are 0.0194 (double well), 0.00485 (MoG40), 0.00143 (MB), and 0.0141 ( $\phi^4$ ); the corresponding double-well and MoG40 random-atom values are 0.0273 and 0.0164. Energy split- $\hat{R}$  is below 1.11 for the tabulated methods, but this does not extend to every basin or collective-variable trace. For example, maximum values are 1.70 for MoG40 LSC-CP, 1.61 for MoG40 LSC-CP-RA, 1.28 for MoG40 PT, and 1.13 for  $\phi^4$  PT. We therefore use these quantities as finite-horizon diagnostics and interpret them jointly with population errors.

Three conclusions follow from Table 3.

(i) *Between-basin ESS separates the methods qualitatively.* The three local integrators record *exactly zero* worst-basin ESS on every system because at least one basin indicator is constant or unvisited in the recorded traces. This is not equivalent to zero transitions: on MoG40, for example, ULA, MALA, and BAOAB record 222, 230, and 60 label changes, and the corresponding MB counts are 209, 194, and 43. Their large energy ESS values illustrate why energy decorrelation alone is insufficient evidence of global mixing. LSC-CP sustains worst-basin ESS between 229 and 1,952—

| Method                | worst-basin ESS (of 32,000) |       |       |          | energy ESS / signed energy bias |                 |                  |                 |
|-----------------------|-----------------------------|-------|-------|----------|---------------------------------|-----------------|------------------|-----------------|
|                       | DW                          | MoG40 | MB    | $\phi^4$ | DW                              | MoG40           | MB               | $\phi^4$        |
| ULA                   | 0                           | 0     | 0     | 0        | 14,815 / +0.0003                | 707 / +0.001    | 22,223 / +0.007  | 17,212 / +0.133 |
| MALA                  | 0                           | 0     | 0     | 0        | 14,869 / +0.0005                | 710 / -0.004    | 21,994 / -0.001  | 15,250 / -0.010 |
| BAOAB                 | 0                           | 0     | 0     | 0        | 3,658 / -0.0002                 | 534 / -0.001    | 5,622 / +0.004   | 2,734 / +0.028  |
| PT                    | 1,211                       | 572   | 2,336 | 2,567    | 20,494 / -0.001                 | 1,340 / +0.0006 | 25,202 / -0.0004 | 24,383 / -0.002 |
| LSC-CP <sup>(*)</sup> | 1,648                       | 229   | 1,952 | 1,150    | 31,891 / +0.194                 | 22,718 / +0.114 | 26,720 / +0.039  | 30,972 / +0.251 |
| LSC-CP-RA             | 1,809                       | 59    | —     | —        | 31,224 / +0.247                 | 9,165 / +0.448  | —                | —               |

Table 3: Post-settling trace diagnostics aggregated over all seeds and chains. Worst-basin ESS is the minimum, over basins, of the summed per-chain effective sample size of that basin’s indicator (Geyer IAT; constant chains contribute zero). Energy bias is the signed difference between the trace mean of  $V$  and the reference target, in energy units; it is interpreted with its observable-specific Monte Carlo standard error. <sup>(\*)</sup>LSC-CP row: deterministic/semi-analytic score on DW and MoG40, paired multi-atom on MB and  $\phi^4$ . Split- $\hat{R}$  and MCSE values for every observable are reported in the supporting data.

the same order as fully costed parallel tempering—using local gradients and the jump-bank score.

(ii) *Mixing and accuracy must be read together.* On MoG40 and  $\phi^4$ , PT has the larger post-settling worst-basin ESS (572 versus 229; 2,567 versus 1,150), whereas LSC-CP has smaller terminal basin KL at the reported horizon (Table 2). The  $\phi^4$  population difference is reference-sensitive as discussed above. Neither diagnostic alone ranks the samplers; population error, observable bias, mixing, and computational cost must be reported together.

(iii) *The LSC chains carry a measurable energy bias.* The post-settling energy mean of the LSC chains exceeds the target by +0.04 to +0.25 energy units, and the MCSE values above resolve these differences. The local methods are not uniformly unbiased either: on  $\phi^4$ , the signed biases are +0.1334 for ULA, -0.0100 for MALA, and +0.0276 for BAOAB. The LSC bias cannot be attributed merely to high-energy post-jump chord configurations, because those transients also occur in the exact continuous-time process whose drift-jump balance preserves  $\pi_\beta$ . Instead, it indicates observable-dependent invariant-measure error in the finite-step kernel, potentially from Euler splitting, drift taming, finite quadrature, exponent capping, or box projection. These devices are useful for robust computation, but systematic refinement is needed to separate their contributions to the measured bias.

*Cost-normalized throughput.* Per-second comparisons are restricted to MB and  $\phi^4$ , which were timed on a dedicated H200 GPU. On MB, LSC-CP and PT yield respectively 33.03 and 48.23 energy ESS/s, while their worst-basin throughputs are 2.41 and 4.47 ESS/s. LSC-CP evaluates  $A_{q\theta} = 4 \times 8 = 32$  score-quadrature energy differences per particle step. On  $\phi^4$ , the energy throughputs are 50.80 and 62.17 ESS/s and the worst-basin throughputs are 1.89 and 6.55 ESS/s, with  $A_{q\theta} = 8 \times 32 = 256$  for LSC-CP. Per-gradient, per-potential, and per-score-quadrature ESS are included for every observable in the supporting data; accounting conventions are given (Appendix F).

## 6 Scope and limitations

Three boundaries of the present evidence are stated explicitly. First, all results are equilibrium-sampling diagnostics of algorithmic chains; passage or round-trip statistics, where recorded, are Kaplan–Meier restricted means of algorithmic mixing times and carry no molecular-kinetics meaning. Second, the finite-step LSC chains are not exactly  $\pi_\beta$ -invariant, and the measured consequences—the energy bias of Section 5 and the clipping/boundary diagnostics of Appendix E—

are part of the reported results. Third, the present comparison does not include a matched-length *random-geometry* jump control (same atom count, length multiset, weights, rate, and cost; randomized directions) that would isolate the contribution of the designed bank geometry from the generic value of long jumps; raw CP controls for the score correction, not for geometry. Box sensitivity has been checked only for one  $\phi^4$  change ( $L = 2$  to  $L = 5$ ); broader checks at  $L = 5, 6, 7$  are needed. Finally, the SNIS and long-PT  $\phi^4$  references differ by basin TV 0.042, so population-error distinctions below that scale are reference sensitive.

## A The Lévy score and the paired-bank estimators

For a finite jump measure  $\nu$ , the jump generator is  $(J_\nu f)(x) = \int [f(x+r) - f(x)] \nu(dr)$ , and the Lévy score is

$$S_{\nu,\beta}(x) = - \int_{\mathbb{R}^d \setminus \{0\}} r \int_0^1 \frac{\pi_\beta(x - \theta r)}{\pi_\beta(x)} d\theta \nu(dr) = - \int r \int_0^1 e^{\beta[V(x) - V(x - \theta r)]} d\theta \nu(dr). \quad (3)$$

The chord fundamental theorem of calculus gives, for smooth compactly supported  $f$ ,

$$\int S_{\nu,\beta}(x) \cdot \nabla f(x) \pi_\beta(x) dx = - \int (J_\nu f)(y) \pi_\beta(y) dy, \quad (4)$$

hence  $\int A_{\nu,\beta} f \pi_\beta dx = 0$  for the generator (2): an infinitesimal, continuous-time weak stationarity statement.

For a finite bank  $\nu_{\mathcal{R}}(dr) = \lambda \sum_{a=1}^A w_a \delta_{R_a}(dr)$  the score is

$$S_{\mathcal{R},\beta}(x) = -\lambda \sum_{a=1}^A w_a R_a I_a(x), \quad I_a(x) = \int_0^1 e^{\beta[V(x) - V(x - \theta R_a)]} d\theta, \quad (5)$$

with the chord integral evaluated by Gauss–Legendre quadrature. For the deterministic shell calculation, radial quadrature additionally replaces the continuous shell law by a finite quadrature measure; the full-law jump step still samples the continuous shell. This is a separate approximation from time discretization. For MoG40, the chord and radial integrals are analytic, while the angular integral uses a finite trapezoidal rule.

The score is accumulated in log space. For each particle, the largest log magnitude is extracted over the positive quadrature/atom contributions *before* their signed vector sum is formed. Rescaling by this common maximum preserves cancellation among opposite vectors. The implementation then replaces the common magnitude  $\exp(M)$  by  $\exp[\min(M, M_{\max})]$ , with  $M_{\max} = 600$ . When active, this cap preserves the relative atom contributions and the score direction but changes its overall magnitude. It is therefore a numerical-stability device, not an algebraically exact score evaluation. Its activation frequency and the maximum pre-cap exponent are recorded in Appendix E.

### A.1 Paired random banks

When the intended law is a mixture of conditional atom laws,  $\nu(dr) = \lambda \sum_a w_a q_a(dr)$ , the paired multi-atom estimator draws, at the start of each step and independently of the state,

$$R_{n,a} \sim q_a, \quad \widehat{\nu}_n(dr) = \lambda \sum_{a=1}^A w_a \delta_{R_{n,a}}(dr), \quad \mathbb{E}[\widehat{\nu}_n] = \nu, \quad (6)$$

evaluates a finite- $q_\theta$  approximation to (5) for the *realized* bank, and draws the jump counts conditionally on that same bank,

$$N_{n,a} \sim \text{Poisson}(\lambda w_a \Delta t), \quad \Delta J_n = \sum_{a=1}^A N_{n,a} R_{n,a}. \quad (7)$$

With the exact chord integral, each frozen bank would satisfy the weak drift–jump identity (4). The finite chord rule makes this identity approximate, but pairing still removes the additional mismatch that would arise if the score and jumps used independent banks. Averaging over the atom draw

makes the estimator unbiased for the *quadrature-defined* score, not for the exact continuous-score integral. The single-atom variant LSC-CP-RA is the  $A = 1$  case with the atom family resampled each step.

The implemented step, with taming scale  $c$  set to twice the shell half-width and integration box  $\mathcal{B}$ , is

$$X_{n+1} = \text{clip}_{\mathcal{B}} \left[ X_n + \Delta t \frac{b_n}{1 + \Delta t \|b_n\|/c} + \sqrt{2\Delta t/\beta} \xi_n + \sum_a N_{n,a} R_{n,a} \right], \quad (8)$$

$$b_n = -\nabla V(X_n) + S_{\mathcal{R}_n, \beta}(X_n).$$

Euler splitting, finite quadrature, exponent capping, drift taming, and box projection all lie outside the exact generator identity. The weak-identity residual tests the uncapped quadrature-level drift-jump cancellation only; it does not test these finite-step modifications. The atomwise counts in (7) are untruncated. The separate full-law compound-Poisson implementation uses a safety cap  $K_{\max} = 8$  on applied multiplicity while recording the uncapped Poisson draw; no cap hit occurred in the reported trajectories. Multiple events for one paired atom within a step reuse its stored displacement, an  $O(\Delta t^2)$  difference from independently resampling every full-law event.

## B System specifications

**Double well.**  $V(x) = (x^2 - 1)^2$ ,  $\beta = 8$ ; bank: atoms  $\pm 2$ , weights  $1/2$ , radial half-width  $0.2$ ; box  $[-5.2, 5.2]$  (widened so that raw CP’s excursions do not pile mass at the edge of the density grid).

**MoG40.** Means  $\mu_k$  drawn once (fixed seed) in  $[-40, 40]^2$ ;  $\pi_\beta = \frac{1}{40} \sum_k N(\mu_k, I_2)$ ; annulus law with radius uniform on  $[4, 15]$ , direction uniform; box  $[-65, 65]^2$ .

**Depth-balanced Müller–Brown.** Latent potential

$$V_{\text{MB3}}(z) = \sum_{k=1}^4 D_k \exp\{a_k(z_1 - x_k)^2 + b_k(z_1 - x_k)(z_2 - y_k) + c_k(z_2 - y_k)^2\}, \quad (9)$$

| $k$ | $D_k$   | $a_k$ | $b_k$ | $c_k$ | $x_k$ | $y_k$ |
|-----|---------|-------|-------|-------|-------|-------|
| 1   | -1.6607 | -1    | 0     | -10   | 1     | 0     |
| 2   | -1.0000 | -1    | 0     | -10   | 0     | 0.5   |
| 3   | -1.0218 | -6.5  | 11    | -6.5  | -0.5  | 1.5   |
| 4   | 0.1500  | 0.7   | 0.6   | 0.7   | -1    | 1     |

(depths retuned so all three minima sit at  $V \simeq -0.7957$ ), embedded as  $x = zB^\top$  with eight auxiliary coordinates of scale  $\sigma_{\text{aux}} = 0.4$ ;  $\beta = 24$ . Bank: four directed relay atoms  $A \rightarrow B$ ,  $B \rightarrow A$ ,  $B \rightarrow C$ ,  $C \rightarrow B$ , weights  $1/4$ , radial half-width  $0.1 \min_a \|r_a\|$ ; drift cap twice that half-width.

**Coupled  $\phi^4$ .** Site potential

$$W(u, v) = (u^2 - 1)^2 + (v^2 - 1)^2 - 0.0125 uv + 0.0075 u + 0.015 v, \quad (10)$$

$\kappa = 2.5$ ,  $\delta = 1/12$ , twelve periodic sites,  $\beta = 8$ . Bank: eight coherent edge atoms  $\mathbf{1}_{12} \otimes (v_j - v_i)$  along the phase square, weights  $1/8$ , radial half-width  $0.1 \min_a \|r_a\|$ , drift cap twice that half-width.

The integration box half-width  $L = 5$  is *derived*: a simultaneous Laplace-component envelope with union-bound tail budget  $\alpha = 10^{-8}$  gives  $B_\beta = 1.910$ , the hottest-PT envelope  $B_{\beta_{\min}=1} = 3.569$ , and the maximum componentwise one-jump reach  $R_\infty = 2.200$ ;  $L = \lceil \max\{B_\beta + R_\infty, B_{\beta_{\min}}\} \rceil = 5$ . The basin map for  $\bar{q}$  is constructed by gradient flow on  $[-4, 4]^2$ —the *jump-reachable* order-parameter set (double-jump reach  $|\bar{q}| \leq 3.3$ ), not merely the phase minima (Appendix E).

**Parallel-tempering ladders (auto-tuned, fully costed).** Double well:  $\beta \in \{8, 1\}$  ( $K = 2$ , swap acceptance 0.41); MoG40:  $\{8, 0.447, 0.025\}$  ( $K = 3$ , 0.26); MB:  $\{24, 13.5, 7.6, 4.3\}$  ( $K = 4$ , 0.37);  $\phi^4$ : six geometric rungs from 8 to 1 ( $K = 6$ , 0.31).

## C Metric definitions

**Basin occupancy, TV, and KL.** For basin partition  $\{\Omega_k\}_{k=1}^K$  with target masses  $p^\star$ ,  $\hat{p}_k = \frac{1}{N} \sum_i \mathbf{1}\{X_i \in \Omega_k\}$ ,  $\text{TV}_{\text{basin}} = \frac{1}{2} \sum_k |\hat{p}_k - p_k^\star|$ , and

$$D_{\text{KL}}(p^\star \|\tilde{p}) = \sum_{k:p_k^\star > 0} p_k^\star \log \frac{p_k^\star}{\tilde{p}_k}, \quad \tilde{p}_k = \frac{\hat{p}_k + \alpha/N}{1 + K\alpha/N}, \quad \alpha = \frac{1}{2}. \quad (11)$$

The target-to-empirical orientation penalizes a sampler that misses a target basin; the Jeffreys-scale pseudocount (reported) keeps the statistic finite.

**Free-energy-surface RMSE.** On a fixed rectangular grid with cell probabilities  $p_i$  and volumes  $|C_i|$ , the reduced free energy is  $A_i = -\log(p_i/|C_i|)$ . With  $\Delta_i = \hat{A}_i - A_i^\star$  on the supported cells  $\mathcal{I} = \{i : p_i^\star \geq 5/N\}$  and reference weights  $w_i = p_i^\star$ ,

$$\text{RMSE}_{F/k_B T} = \left[ \frac{\sum_{\mathcal{I}} w_i (\Delta_i - c^\star)^2}{\sum_{\mathcal{I}} w_i} \right]^{1/2}, \quad c^\star = \frac{\sum_{\mathcal{I}} w_i \Delta_i}{\sum_{\mathcal{I}} w_i}, \quad (12)$$

i.e. free energies are compared after fitting the single additive constant they are defined up to. Grids use the two-dimensional active coordinates (one-dimensional for the double well), one-half pseudocount per cell, and the empirical mass outside the frozen grid is reported separately (Table 2) rather than renormalized away.

**IAT, ESS, MCSE,  $\hat{R}$ .** Per chain, the integrated autocorrelation time uses Geyer’s initial-positive/initial-monotone pairwise sums; a constant chain is assigned  $\tau_{\text{int}} = \infty$  and zero ESS. Independent chains contribute  $\text{ESS}_{\text{total}} = \sum_m n_m / \widehat{\tau_{\text{int}m}}$  (not draws over mean IAT).  $\text{MCSE} = s/\sqrt{\text{ESS}}$ , undefined for zero-variance traces. Split- $\hat{R}$  is rank normalized with average ranks for ties and Blom constant  $(r-3/8)/(S+1/4)$ , and the reported value is the maximum of the bulk and folded (median-absolute-deviation) statistics; globally constant traces give an undefined  $\hat{R}$ . Cost-normalized ESS divides by wall time, gradient evaluations, potential evaluations, and score-quadrature energy differences separately; observation functions run outside both the timer and the evaluation ledger.

**Passage statistics.** Where passage times are recorded they are right-censored at the common horizon, restricted to chains starting in the declared home core, and summarized by the Kaplan–Meier restricted mean; the model-dependent censored-exponential estimate, when saved, is labelled as such. Both are algorithmic-mixing diagnostics only.

## D References and the $\phi^4$ importance reference

Double well: dense-grid inverse CDF on the declared box ( $2 \times 10^5$  grid points). MoG40: exact mixture draws. Müller–Brown: fine latent grid ( $2400^2$ ) times exact Gaussian auxiliaries.  $\phi^4$ : Laplace mixture at the four coherent minima (autograd Hessians) corrected by self-normalized importance weights. Weight diagnostics:  $2 \times 10^5$  proposals, weight ESS 41,182 (20.6%), entropy ESS consistent, maximum normalized weight  $3.6 \times 10^{-3}$ ; basin masses, energy moments, CV means, and the FES reference histogram use the weights directly (no resampling); finite SIR supplies unweighted clouds only where required ( $W_2$ , MMD) and is labelled approximate. Reference floors in Table 2 are two-draw floors of the corresponding reference at matched sample size.

## E Numerical convergence and stability diagnostics

| Diagnostic                               | DW                    | MoG40                  | MB                     | $\phi^4$                     |
|------------------------------------------|-----------------------|------------------------|------------------------|------------------------------|
| nonfinite states                         | 0                     | 0                      | 0                      | 0                            |
| score-cap fraction, reported LSC         | $5.02 \times 10^{-4}$ | 0                      | $2.85 \times 10^{-7}$  | $2.09 \times 10^{-4}$        |
| state-box fraction, maximum over methods | $8.67 \times 10^{-5}$ | $3.84 \times 10^{-4}$  | $3.304 \times 10^{-3}$ | $8.903 \times 10^{-4}$       |
| jump-boundary contact, reported LSC      | $1.69 \times 10^{-4}$ | $1.05 \times 10^{-5}$  | $2.18 \times 10^{-3}$  | $1.60 \times 10^{-4}$        |
| jump-boundary contact, raw CP            | $3.03 \times 10^{-4}$ | $2.475 \times 10^{-2}$ | $6.733 \times 10^{-2}$ | $4.130 \times 10^{-3}$       |
| basin-map escape, targeting methods      | 0                     | 0                      | 0                      | exactly $1.0 \times 10^{-3}$ |
| uncapped weak-identity residual          | $1.5 \times 10^{-7}$  | $3.0 \times 10^{-8}$   | $5.0 \times 10^{-7}$   | $2.0 \times 10^{-7}$         |

Table 4: Numerical-stability measurements for the reported trajectories. Score-cap and state-box entries are fractions of particle proposals; the state-box row is the maximum over all compared methods. Jump-boundary contact is measured per applied jump. The  $\phi^4$  basin-map entry corresponds to one particle at one checkpoint ( $1/N$  with  $N = 1000$ ).

The score-cap activation is small but not identically zero for the double well, MB, or  $\phi^4$ . On the deliberately extended weak-identity grids, the fractions with pre-cap exponent above  $M_{\max}$  are 0.402 for the double well and 0.374 for MB; these grid values are distinct from the trajectory activation frequencies in Table 4. The residual itself is formed from the uncapped log-domain score and tests the generator identity for the quadrature-defined measure and a finite family of observables. It does not test the exponent cap, drift taming, Euler splitting, or box projection, and it is not a certificate of the invariant distribution of the stabilized finite-step chain.

Because compound-Poisson jumps occur independently of the current state, rare events can contact any finite computational box. The contact and projection frequencies in Table 4 quantify this exposure. In particular, the larger raw-CP contact rates on MoG40 and MB make the finite box a material confound in that ablation. For  $\phi^4$ , the basin map was constructed on  $[-4, 4]^2$  to cover the jump-reachable order-parameter region rather than only the phase minima. Even rare cap or projection events alter the numerical kernel, so low frequencies are evidence of numerical robustness, not exact stationarity.

## F Computational details and data availability

All calculations used float64 arithmetic on one NVIDIA H200 GPU, with Python 3.12.13 and PyTorch 2.13.0+cu130. Complete seed lists, numerical configurations, raw and time-series metrics,

analysis-ready summaries, and stationarity traces accompany the supporting data. Gradient, potential, and Lévy-score-quadrature evaluation counts are stored separately for each method and observable.

The MB and  $\phi^4$  ESS/s values in Section 5 were measured without a concurrent GPU workload; for the double well and MoG40, we report hardware-independent evaluation counts rather than wall-clock-normalized efficiency. Source code, configurations, and analysis-ready numerical artifacts will be deposited in a public archival repository upon publication.